

ARTIFICIAL INTELLIGENCE INTERNSHIP

Stock Market Prediction Using Machine Learning

Internship Project Presentation

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Codec Technologies Pvt. Ltd. | AI Intern | 02 Jun 2026 – 02 Jul 2026

About the Organization

Codec Technologies Pvt. Ltd.

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Internship Details

DESIGNATION

Artificial Intelligence Intern

MODE

Pan India, Hybrid

DURATION

02/06/2026 - 02/07/2026

REPORTING TO

Assigned Project Head(s)

SUPERVISOR

Dr. Anurag Shrivastava
Talent Acquisition Manager

Problem Statement & Objectives

The Problem

Stock prices are highly volatile and non-linear, shaped by company performance, market sentiment, and macroeconomic factors. Traditional statistical methods often fail to capture these complex patterns — creating a need for machine learning models that can learn from historical trends.

Goal

Build a model that analyses historical price data and generates reasonably accurate short-term predictions to assist in understanding market behaviour.

Objectives

- 1 Understand factors driving stock price movement
- 2 Collect and preprocess historical price data
- 3 Perform exploratory data analysis & visualization
- 4 Build regression + LSTM time-series models
- 5 Evaluate performance using RMSE / MAE
- 6 Gain hands-on, industry-relevant AI/ML experience

Technologies & Tools Used

1

Python 3

Core programming language for the entire pipeline

2

Pandas & NumPy

Data handling, cleaning, and preprocessing

3

Matplotlib & Seaborn

Exploratory data analysis and visualization

4

Scikit-learn

Baseline Linear Regression model & evaluation metrics

5

TensorFlow / Keras

LSTM-based deep learning time-series model

6

Jupyter / Google Colab

Development and experimentation environment

Methodology & Workflow

1

Data Collection

Historical daily OHLCV stock data gathered

2

Preprocessing

Cleaning, scaling & sequence creation (60-day window)

3

EDA

Trend, moving average & volatility visualization

4

Model Building

Linear Regression baseline + LSTM network

5

Training & Validation

Train/test split, Adam optimizer, MSE loss

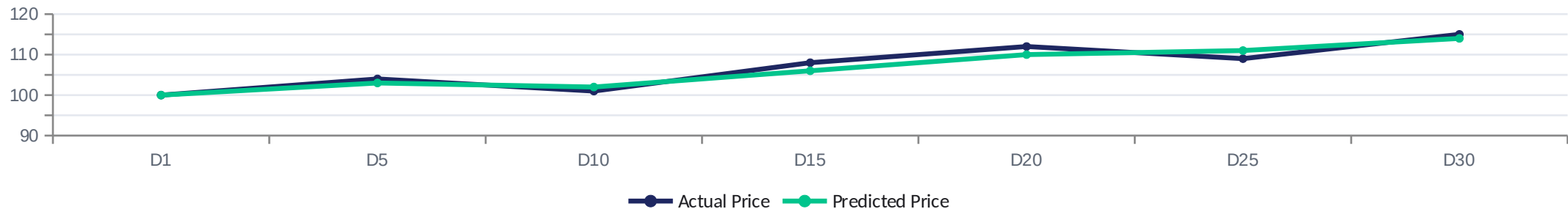
6

Evaluation

RMSE / MAE + predicted vs. actual price plots

The LSTM (Long Short-Term Memory) network was chosen for its ability to capture temporal dependencies in sequential financial data, making it well suited for time-series forecasting tasks such as stock price prediction.

Illustrative Output — Predicted vs. Actual Closing Price Trend



Outcomes & Key Learnings

1

Working Prediction Model

A trained ML model forecasting short-term stock closing prices from historical trends.

2

Visual Trend Comparison

Plots comparing predicted vs. actual prices to assess model behaviour.

3

Quantified Accuracy

Model evaluated using RMSE and MAE error metrics.

4

Applied AI/ML Skills

Practical, industry-mentored experience in the end-to-end ML workflow.

Internship Completion



Verified Credential

AWARDED TO

Arthav Kumar Som

ROLE

Artificial Intelligence Intern

DURATION

02/06/2026 - 02/07/2026

ISSUED BY

Dr. Anurag Shrivastava, Program Manager

PROGRAM

AICTE & ICAC Approved

Thank You

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